



Titouan Guerin

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M2 Machine Learning student at Sorbonne Université specializing in Deep Reinforcement Learning and World Models. Looking to pursue research through a PhD.

Education

Sorbonne Université, MSc in AI & Machine Learning Sept 2023 – Present

- Specialization: Probabilistic Modeling, Machine Learning & Big Data, Mathematics and Research focus
- **Coursework:** Statistics, Deep Learning, NLP, LLM, Reinforcement Learning...
- **Class Rank:** 8/40

University of Luxembourg, BS in Computer Science Sept 2020 – Aug 2023

- Member of the Competitive Programming Club (participated in BAPC 2021)
- **Coursework:** Linear Algebra, Algorithms & Complexity, Advanced Programming, Data Science ...

Experience

Research Internship, ISIR x ONERA – Paris March 2026 – Aug 2026

- Research on world models and deep reinforcement learning for drone control
- **Key-words:** Physics informed world models, model-based reinforcement learning, model predictive control

AI Research Assistant, AI Robolab – Luxembourg July 2024 – Aug 2024

- Lead backend researcher for a multi-agent system learning app powered by LLMs using LangGraph, designed for low-resource language learning (Luxembourgish).
- Co-authored a peer-reviewed paper presented at PRIMA conference in Kyoto, Japan.

Publications

- Nouzri, S., El Fatimi, M., **Guerin, T.**, Othmane, M., Najjar, A. (2025). *Beyond Chatbots: Enhancing Luxembourgish Language Learning Through Multi-agent Systems and Large Language Model*. In: *PRIMA 2024: Principles and Practice of Multi-Agent Systems*, Kyoto, Japan. Lecture Notes in Computer Science, vol. 15395. Springer, Cham. https://doi.org/10.1007/978-3-031-77367-9_29

Projects

Reinforcement Learning for Dark Souls – [link](#)

- Implemented model-free RL algorithms from scratch to train an agent in *Dark Souls III* using the SoulsGym environment (custom Gymnasium).
- Frameworks: Gymnasium, Pytorch, Tensorboard

From Scratch Neural Networks – [link](#)

- Implemented Neural Networks from scratch (using only numpy) in a modular way
- Features: Activation Functions, Linear/Pooling/Conv Layers, Loss Functions, SGD Optimizer, PCA, K-Means

Skills

Programming: Python, Javascript, SQL, Bash

ML & Data: PyTorch, TensorFlow, scikit-learn, Gymnasium, TorchRL, NumPy, Pandas, TensorBoard

Development Tools: Git, FastAPI, React, Docker, Linux

Languages: English/French (Bilingual), German (B1)